

Quartic Zero-Repair: Final Phase-Lift Closure Target

1 Exact model

Work with

$$P = \frac{1}{64}(8, 16, 16, 16, 8), \quad Q = \frac{1}{64}(7, 20, 10, 20, 7)$$

on $\{0, 1, 2, 3, 4\}$.

For a probability law

$$B(z) = \sum_{j \geq 0} b_j z^j$$

an exact zero-repair fan consists of probability laws $B_0, \dots, B_4$ satisfying

$$\boxed{Q(z)B(z) = \sum_{x=0}^4 P_x z^x B_x(z).} \quad (\text{ZR})$$

The exact Laurent bulk factors are

$$K_0(z) = \frac{7+z}{8}, \quad K_1(z) = 1,$$

$$K_2(z) = \frac{3z^{-1} + 10 + 3z}{16}, \quad K_3(z) = 1, \quad K_4(z) = \frac{z^{-1} + 7}{8},$$

and satisfy

$$\sum_{x=0}^4 P_x z^x K_x(z) = Q(z).$$

We henceforth work on the forward-invariant face

$$\boxed{b_0 = 0.}$$

On this face branches 1 and 3 may be taken identically:

$$B_1 = B_3 = B.$$

2 Exact phase parametrization

Let

$$A := BK_0.$$

Choose positive donor measures τ_2, τ_4 with

$$|\tau_2| = \frac{3b_1}{8}, \quad |\tau_4| = \frac{b_1}{8},$$

such that

$$\tau_2 + \tau_4 \leq A.$$

Their corresponding positive reinjections are

$$\mu_2 = \frac{1}{2}z^{-2}\tau_2, \quad \mu_4 = z^{-4}\tau_4.$$

Then define

$$B_2 = BK_2 - \frac{3b_1}{16}\delta_0 + \mu_2, \tag{P2}$$

$$B_4 = BK_4 - \frac{b_1}{8}\delta_0 + \mu_4, \tag{P4}$$

and

$$B_0 = BK_0 + \frac{3b_1}{8}z^2 + \frac{b_1}{8}z^4 - \tau_2 - \tau_4. \tag{P0}$$

These five children satisfy *the same single parent fan* and therefore the same prefix decision. In particular, this is not a wordwise post-selection construction.

Lemma 1 (Exact sibling compensation). *Whenever the right-hand sides of (P0)–(P4) are nonnegative probability laws, they satisfy (ZR) exactly.*

Proof. The branch-2 correction contributes

$$P_2 z^2 \left(-\frac{3b_1}{16}\delta_0 + \mu_2 \right),$$

while the corresponding branch-0 compensation is

$$P_0 \left(\frac{3b_1}{8}z^2 - \tau_2 \right).$$

Since

$$P_0 = \frac{1}{8}, \quad P_2 = \frac{1}{4}, \quad \tau_2 = 2z^2\mu_2,$$

the two contributions cancel identically.

Likewise

$$P_0 \left(\frac{b_1}{8}z^4 - \tau_4 \right) + P_4 z^4 \left(-\frac{b_1}{8}\delta_0 + \mu_4 \right) = 0$$

because $P_0 = P_4 = 1/8$ and $\tau_4 = z^4\mu_4$. The untouched bulk factors already satisfy the bulk identity. $\square$

3 Branch 4 can be frozen

Let

$$\mathcal{M} := \{B : b_0 = 0, \quad b_1 \geq b_2 \geq b_3 \geq \dots \geq 0\}.$$

For $B \in \mathcal{M}$ define

$$\mu_4^B(j) = \frac{b_j - b_{j+1}}{8}, \quad j \geq 1, \tag{S4}$$

and

$$\tau_4^B = z^4\mu_4^B.$$

Then

$$|\mu_4^B| = \frac{b_1}{8},$$

and direct coefficient comparison gives

$$B_4 = B. \tag{Id4}$$

Thus branches 1, 3, 4 need no phase evolution. Only the sibling pair (0, 2) remains dynamically nontrivial.

4 Exact one-step Hall region

Define

$$\mathcal{R} = \{B : b_0 = 0, \quad 12b_1 + 7b_2 \leq 8, \quad 9b_1 + 8b_2 + 8b_3 + 7b_4 \leq 8\}. \quad (\text{R})$$

Lemma 2 (Exact donor Hall criterion). *Let B satisfy $b_0 = 0$ and let $A = BK_0$. There exist disjoint positive donor packets*

$$\tau_2, \tau_4 \leq A,$$

with

$$|\tau_2| = \frac{3b_1}{8}, \quad |\tau_4| = \frac{b_1}{8},$$

and supports

$$\tau_2 \subseteq \{3, 4, \dots\}, \quad \tau_4 \subseteq \{5, 6, \dots\},$$

if and only if

$$A([3, \infty)) \geq \frac{b_1}{2}, \quad A([5, \infty)) \geq \frac{b_1}{8}.$$

Equivalently,

$$12b_1 + 7b_2 \leq 8,$$

and

$$9b_1 + 8b_2 + 8b_3 + 7b_4 \leq 8.$$

Hence $\mathcal{R}$ is exactly the one-step positive Hall region of the minimum-row phase grammar.

5 Weak- L^1 seed

Define

$$H_2(j) = \frac{2}{(j+1)(j+2)}, \quad j \geq 1, \quad H_2(0) = 0. \quad (\text{H2})$$

Then

$$\sum_{j \geq 1} H_2(j) = 1,$$

and

$$\mathbb{P}_{H_2}(J \geq j) = \frac{2}{j+1} \leq \frac{2}{j}. \quad (\text{WL1})$$

Moreover $H_2 \in \mathcal{M} \cap \mathcal{R}$.

An explicit exact first phase step is already available:

$$H_2 \mapsto (F_0, H_2, F_2, H_2, H_2),$$

where one may choose

$$\tau_2 = \frac{13}{120}\delta_3 + \frac{1}{60}\delta_4,$$

equivalently

$$\mu_2 = \frac{13}{240}\delta_1 + \frac{1}{120}\delta_2,$$

while branch 4 is frozen by (S4).

Thus the phase-lift problem is genuinely recursive, not a one-step existence problem.

6 The remaining theorem

For $A < \infty$ define the uniform weak- L^1 cone

$$\mathcal{W}_A = \left\{ B : \mathbb{P}_B(J \geq j) \leq \frac{A}{j} \text{ for every } j \geq 1 \right\}.$$

Theorem 3 (Phase-Lift Closure Lemma — FINAL TARGET). *There exist a finite constant A and a nonempty countable phase class*

$$\boxed{\mathcal{V} \subseteq \mathcal{M} \cap \mathcal{R} \cap \mathcal{W}_A}$$

such that

$$H_2 \in \mathcal{V},$$

and such that for every $B \in \mathcal{V}$ there exists a positive donor selection

$$\tau_B$$

with

$$|\tau_B| = \frac{3b_1}{8}, \quad \tau_B \subseteq \{3, 4, \dots\},$$

for which, after defining the stationary branch-4 donor by (S4),

$$\tau_4^B = z^4 \mu_4^B,$$

we have

$$\tau_B + \tau_4^B \leq BK_0,$$

and the two nontrivial children

$$\mathsf{T}_0(B; \tau_B) := BK_0 + \frac{3b_1}{8}z^2 + \frac{b_1}{8}z^4 - \tau_B - \tau_4^B, \quad (1)$$

$$\mathsf{T}_2(B; \tau_B) := BK_2 - \frac{3b_1}{16}\delta_0 + \frac{1}{2}z^{-2}\tau_B \quad (2)$$

both satisfy

$$\boxed{\mathsf{T}_0(B; \tau_B) \in \mathcal{V}, \quad \mathsf{T}_2(B; \tau_B) \in \mathcal{V}.} \quad (\text{PL})$$

The donor selection must be determined from the current parent phase B alone, before the next public branch is observed.

Remark 4 (What counts as a proof). *A proof of Theorem 3 must exhibit the class $\mathcal{V}$ and a constructive parent-measurable rule $B \mapsto \tau_B$, or an equivalent positive existence argument. Numerical LP feasibility, finite-depth verification, separate wordwise choices, or a branch-dependent choice made after observing the child do not prove the theorem.*

7 Why this closes the problem

Theorem 5 (Phase lift implies an infinite exact tree). *Assume Theorem 3. Then there exists an exact prefix-causal zero-repair tree of arbitrary depth, indeed an infinite tree, rooted at H_2 .*

Proof. Start from $B_\emptyset = H_2 \in \mathcal{V}$.

At every history h whose state is $B_h \in \mathcal{V}$, choose the single parent donor τ_{B_h} supplied by Theorem 3. Define

$$B_{h0} = \mathsf{T}_0(B_h; \tau_{B_h}), \quad B_{h2} = \mathsf{T}_2(B_h; \tau_{B_h}),$$

and

$$B_{h1} = B_{h3} = B_{h4} = B_h.$$

The stationary branch-4 construction and the exact sibling compensation lemma imply that these five laws form one exact positive ZR fan.

By (PL), all children again lie in $\mathcal{V}$. Induction over ordered histories therefore constructs the full infinite prefix tree. $\square$

Theorem 6 (Logarithmic upper bound). *Assume Theorem 3. Then*

$$\boxed{\mathcal{B}_n = O(\log n).}$$

More explicitly,

$$\boxed{\mathcal{B}_n \leq \mathbb{E}_{H_2} \min(J, 4n) \leq 2H_{4n}.}$$

Proof. The infinite exact tree rooted at H_2 can be truncated at depth n by replacing the initial reserve J with

$$J^{(n)} = \min(J, 4n),$$

since over n steps the reserve can decrease by at most $4n$.

Hence

$$\mathcal{B}_n \leq \mathbb{E} \min(J, 4n) = \sum_{j=1}^{4n} \mathbb{P}(J \geq j).$$

Using

$$\mathbb{P}_{H_2}(J \geq j) = \frac{2}{j+1} \leq \frac{2}{j}$$

gives

$$\mathcal{B}_n \leq 2 \sum_{j=1}^{4n} \frac{1}{j} = 2H_{4n} = O(\log n).$$

$\square$

Corollary 7 (Final asymptotics). *Together with the already proved lower bound*

$$\mathcal{B}_n \geq \kappa_0 \log n - o(\log n), \quad \kappa_0 > 0,$$

Theorem 3 implies

$$\boxed{\mathcal{B}_n = \Theta(\log n).}$$

Astra task

Prove Theorem 3.

The proof should attack the controlled invariance problem

$$B \mapsto (\mathsf{T}_0(B; \tau_B), \mathsf{T}_2(B; \tau_B))$$

rather than re-solving the five-branch tree.

The natural route is to construct a countable positive phase cone containing H_2 in which:

1. the two Hall inequalities defining $\mathcal{R}$ are invariant;
2. the coefficient sequence remains nonincreasing, so branch 4 can be frozen exactly;
3. a uniform weak- L^1 tail bound survives;

4. the same donor packet τ_B simultaneously keeps both T_0 and T_2 inside the cone;
5. donor mass may escape to increasing scales, but the parent rule remains prefix-compatible.

A finite-state stationary closure is neither required nor expected. A countable history-generated phase family is sufficient.

Prove the positive parent-measurable phase selection rule.

Once this is done, no further asymptotic or causal-lift lemma is needed: the infinite exact tree and the logarithmic upper bound follow from the theorems above.